INSIDE A DATA CENTER

A building that coordinates power, computers, storage, networking and people.

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THE SYSTEM AT A GLANCE

01 POWER › 02 COMPUTERS › 03 NETWORK › 04 SERVICE

FOUR IDEAS THAT MATTER

1 RACKS ARE SHELVES

A rack holds multiple servers and network switches. A server is a computer that serves other computers.

2 ALWAYS-ON SUPPORT

Electric power, cooling, fire controls and physical security help the machines operate reliably.

3 SOFTWARE COORDINATES

Programs distribute requests and move work between servers. A user normally sees only the application.

4 NOT A MAGIC CLOUD

The cloud is someone else's real hardware, offered as a service over a network.

MEMORY HOOK

Like a school: rooms, electricity, corridors and schedules support the actual lessons.

DISCUSS WITHOUT COMPUTERS

Which parts still matter if nobody is visiting the website right now?

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ONE CLICK, MANY MACHINES

Follow a request from phone to data center and back.

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THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 REQUEST

Your browser asks for a page or an application result. A request is a structured message.

2 ENTRY POINT

A gateway receives the message. A load balancer may choose a healthy server with available capacity.

3 WORK

The selected service may read a cache, call another service or retrieve records from a database.

4 REPLY

The server sends data back. Your device renders the result and may save a temporary copy.

MEMORY HOOK

The click is local; the computation may travel across many computers.

DISCUSS WITHOUT COMPUTERS

If one service is slow, can the whole request become slow?

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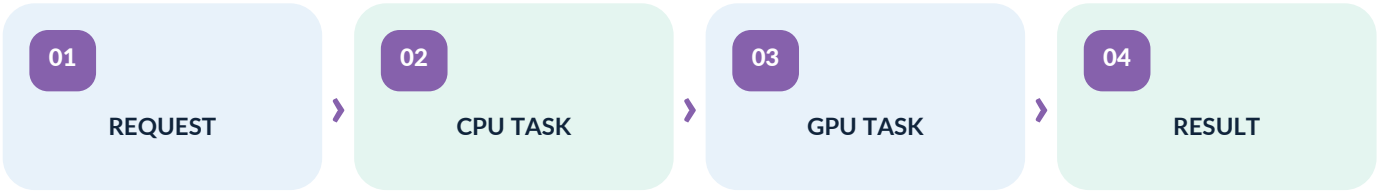
CPU (Central Processing Unit) · GPU (Graphics Processing Unit)

CPUs, GPUs & BATCHES

Different workers, different tasks. Faster hardware is not enough on its own.

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THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 CPU (CENTRAL PROCESSING UNIT)

The general-purpose processor: runs operating-system code, coordinates services and handles many kinds of logic.

2 GPU (GRAPHICS PROCESSING UNIT)

A processor designed for many parallel calculations; especially useful for many matrix operations in AI.

3 BATCHING

Several compatible jobs may be grouped to improve throughput. Waiting for a batch can add latency.

4 LATENCY VS THROUGHPUT

Latency: how long one request waits. Throughput: how many requests finish in a period of time.

MEMORY HOOK

A GPU is not simply a “faster CPU”: it is designed for different patterns of work.

DISCUSS WITHOUT COMPUTERS

Could a full batch improve efficiency but delay one urgent request?

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MEMORY IS NOT STORAGE

Where are active calculations, saved files and backups?

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FOUR IDEAS THAT MATTER

1 RAM (RANDOM ACCESS MEMORY)

Temporary working area for running programs. Its usual contents disappear when power is lost.

2 SSD (SOLID-STATE DRIVE)

Persistent storage that keeps saved information after a normal shutdown.

3 SHARED OR OBJECT STORAGE

Multiple servers can use shared systems. Object storage manages named objects with data and metadata.

4 BACKUP IS DIFFERENT

A replicated copy is not always a backup. A separate, recoverable history protects against deletion and corruption.

MEMORY HOOK

RAM is the worktable. Storage is the cabinet. Backup is the spare copy elsewhere.

DISCUSS WITHOUT COMPUTERS

If a file is deleted and the deletion replicates, is it still backed up?

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AI (Artificial Intelligence)

NETWORKS: THE HALLWAYS

Data centers are as much about movement as they are about computing.

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FOUR IDEAS THAT MATTER

1 ROUTER

Moves network packets between networks, based on network addresses.

2 SWITCH

Connects devices inside a local network and forwards frames to the correct destination.

3 BOTTLENECK

A fast processor may wait for data on a slow network. Delay and congestion matter.

4 AI CLUSTER

AI servers often exchange large amounts of data. Specialized fast links connect accelerator nodes.

MEMORY HOOK

Many fast computers still feel slow when the network becomes a traffic jam.

DISCUSS WITHOUT COMPUTERS

Would adding more servers solve a network bottleneck?

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POWER & COOLING

Every computation becomes heat. Reliability begins before the code runs.

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THE SYSTEM AT A GLANCE

01 GRID › 02 POWER SUPPLY › 03 CHIPS › 04 COOLING

FOUR IDEAS THAT MATTER

1 ELECTRICAL PATH

Power reaches the facility, passes through electrical distribution and supplies computers.

2 UPS (UNINTERRUPTIBLE POWER SUPPLY)

A short-term power system that can bridge interruptions while other protection acts.

3 HEAT HAS TO MOVE

Fans or liquid cooling carry heat away. More computation generally requires more heat removal.

4 EFFICIENCY IS END TO END

Useful work per unit of energy depends on hardware, cooling, utilization and software decisions.

MEMORY HOOK

A server room is a machine whose purpose is to keep other machines usable.

DISCUSS WITHOUT COMPUTERS

What happens to a busy AI cluster if cooling stops?

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AI (Artificial Intelligence)

WHO GETS A SERVER?

The scheduler coordinates requests for finite resources.

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THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 JOB

A job is a piece of work, such as processing a video or running an AI model.

2 SCHEDULER

A program chooses where and when work should run, using rules about capacity and constraints.

3 CAPACITY IS FINITE

Machines have limited processor time, memory and network capacity. Some jobs must wait.

4 FAILURE & RETRY

If a job cannot run, the system may wait and try again. Retries require care to avoid repeated effects.

MEMORY HOOK

Scheduling is like assigning students to a few available lab computers.

DISCUSS WITHOUT COMPUTERS

Should a tiny urgent job always wait behind a very large job?

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SECURITY & FAILURES

Keeping a service working is different from keeping its information safe.

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FOUR IDEAS THAT MATTER

1 AUTHENTICATION

Proves who or what is requesting access. Machines and people may have separate identities.

2 AUTHORIZATION

Decides which action that identity may perform. A reader should not automatically be a writer.

3 ISOLATION

Separate workloads, network zones and credentials limit what an incident can affect.

4 RESILIENCE

Monitoring, multiple copies and tested recovery reduce disruption, but none can prevent every failure.

MEMORY HOOK

Safe infrastructure asks two questions: Who are you? What are you allowed to do?

DISCUSS WITHOUT COMPUTERS

Can a service be online and still leak private records?

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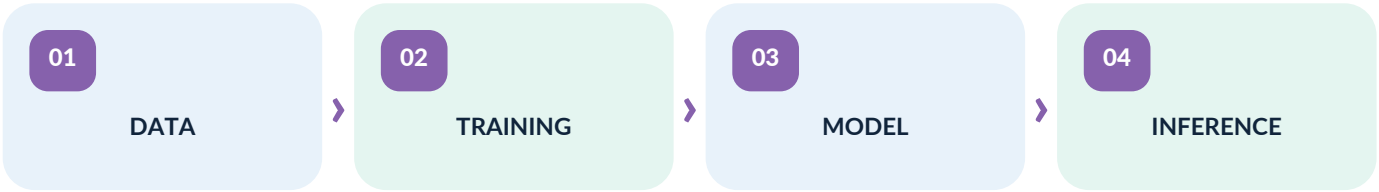
AI (Artificial Intelligence) · KV (Key-Value)

HOW AI CLUSTERS WORK

Training changes model parameters; inference uses a trained model to answer.

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THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 TRAINING

Repeatedly process examples and adjust a model's internal parameters. May use many graphics processors together.

2 CHECKPOINT

Saved training state allows work to resume after interruption. It takes time and storage to save.

3 INFERENCE

Use an existing model to process new input. The serving system manages incoming requests and resources.

4 KV CACHE (KEY-VALUE CACHE)

During some transformer-model inference, stored intermediate attention data helps avoid recomputing earlier tokens.

MEMORY HOOK

Training builds the recipe. Inference uses it. Both need data centers.

DISCUSS WITHOUT COMPUTERS

If the model is already trained, why might a response still be slow?

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FIFO (First In, First Out) · GPU (Graphics Processing Unit)

QUEUES & PRIORITY HEAPS

A structure for waiting can determine who receives expensive hardware next.

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THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 FIFO (FIRST IN, FIRST OUT)

A queue normally removes the oldest item first. This is fair by arrival order, but not always by urgency.

2 PRIORITY QUEUE

Returns the highest-priority item first. The queue can be implemented with a heap (a special tree shape).

3 HEAP (TREE FOR PRIORITY)

A heap lets software find a highest- or lowest-priority item quickly; it does not keep all items fully sorted.

4 THE TRADE-OFF

Strict priority can starve low-priority jobs. Real schedulers may add fairness, quotas or aging rules.

MEMORY HOOK

The scheduler is not “thinking”: it is following ordering and capacity rules.

DISCUSS WITHOUT COMPUTERS

Which structure chooses the oldest job, and which picks the most urgent?

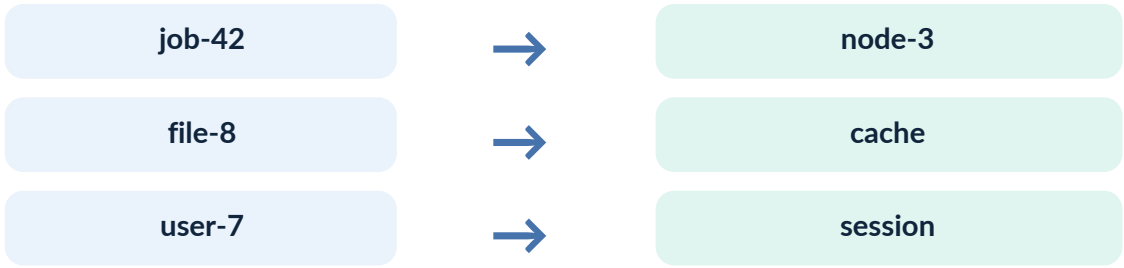
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HASH MAPS, CACHES & SETS

Avoid searching every computer or recalculating the same answer.

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FOUR IDEAS THAT MATTER

1 HASH MAP (KEY → VALUE)

Store an association such as job ID → assigned server. A hash function helps locate the key efficiently.

2 CACHE (FAST COPY)

Keep a recent or popular value near where it is needed. The cache may become outdated.

3 SET (UNIQUE MEMBERS)

Keep unique identifiers: servers already visited or active connections. It prevents duplicate entries.

4 CACHE INVALIDATION

A cached answer is useful only while it is valid for its context. Changes, time and permissions matter.

MEMORY HOOK

A dictionary answers “Where is job 42?”; a cache asks “Can I reuse it safely?”

DISCUSS WITHOUT COMPUTERS

Why might yesterday’s cached inventory give the wrong answer today?

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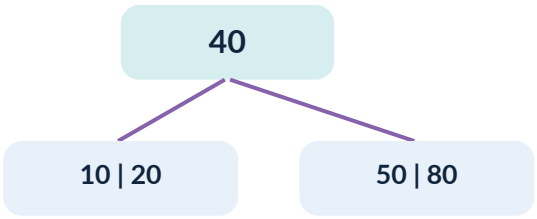
B-tree (balanced search tree index)

TREES, INDEXES & EVENT LOGS

Finding saved data and remembering changes are different operations.

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THE SYSTEM AT A GLANCE



EVENT LOG

- 01 WRITE
- 02 COMMIT
- 03 REPLAY

FOUR IDEAS THAT MATTER

1 B-TREE (BALANCED SEARCH TREE)

A tree-based database index helps find exact or ranged values without examining every record.

2 INDEX IS NOT THE DATA

Like a book index: it points toward records. Indexes improve some reads but add work during writes.

3 APPEND-ONLY LOG

A sequence of events added at the end. Good for following what changed and replaying processing.

4 REPLICATION & ORDER

Several machines may copy logs. Order and delivery guarantees must be defined; “stored” is not always “fully committed”.

MEMORY HOOK

An index finds an item. A log remembers what happened.

DISCUSS WITHOUT COMPUTERS

Why can adding more indexes slow down database updates?

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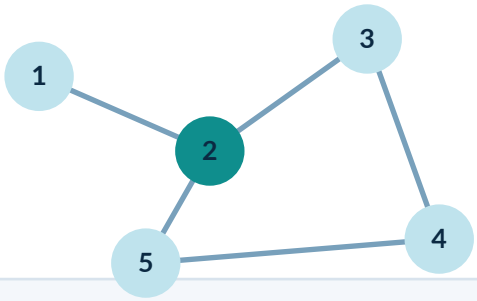
AI (Artificial Intelligence) · RAG (Retrieval-Augmented Generation)

GRAPHS, VECTORS & AI SEARCH

The data structure changes how an AI center finds relevant information.

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THE SYSTEM AT A GLANCE





FOUR IDEAS THAT MATTER

1 GRAPH (NODES + EDGES)

Represents links: service A calls service B; or documents are connected by similarity.

2 VECTOR (LIST OF NUMBERS)

A model can encode a text into numbers called an embedding. Similar meanings may yield nearby vectors.

3 NEAREST-NEIGHBOR INDEX

Special indexes accelerate approximate similarity search; results depend on the index and distance metric.

4 RAG (RETRIEVAL-AUGMENTED GENERATION)

A system retrieves potentially relevant text before generating an answer; retrieval does not guarantee truth.

MEMORY HOOK

A graph shows relationships; a vector index suggests what is similar.

DISCUSS WITHOUT COMPUTERS

Can two documents be numerically similar but have different factual claims?

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